Lesson 1: dates and environmental records

Reading Environment and History through Tree Rings / An introductory course / Handout v5

No advance preparation is required

You do not need to read materials or watch a video before this first class. We will introduce the terms and read the figures together. Open this handout during the session and use it as a reference. We will work through sample A together before you practise with B-D.

Learning goals

1. Explain how tree rings help us understand environment and history.

2. Identify the event that a measured age represents.

3. Suggest a method for a sample and explain the conditions to check.

Terms introduced during class

Term	Meaning in this session
Sample	Material selected for study, such as wood, a seed or a mineral. Check its relationship to the event of interest.
Relative age	The order of events: which came first?
Numerical age	An age expressed as a number or range. Absolute dating also involves uncertainty.
Calendar year	A year on a calendar. Annual steps alone do not establish calendar placement.
Uncertainty	Information about an age range or the strength of an estimate.
Temporal resolution	The time interval that a record can distinguish. It differs from dating accuracy.

Opening question

If someone makes modern furniture from old buried wood, which event would the wood date? Consider ring formation, felling, construction and reuse separately.

Samples and the events we can date

Use this table during the exercise. You do not need to memorise the method names in advance.

Method or approach	Example material	Dated event and checks
Relative dating	Stratigraphic layers	Establishes order. Check overturning, redeposition and mixing.
Dendrochronology	Wood with tree rings	Dates surviving rings. Assess missing outer wood and reuse before interpreting felling or construction.
Radiocarbon	Carbon in charred seeds or wood	Relates to carbon incorporation. Check contamination and context, then calibrate to calendar time.
K-Ar / Ar-Ar	Suitable volcanic minerals	Relates to cooling and gas retention. Check minerals and depositional context.
U-Th	Suitable carbonates, including stalagmites	Dates carbonate formation under suitable conditions. Assess initial material and later exchange.
Luminescence	Sediment or heated material with suitable minerals	Assesses time after signal resetting. In the laboratory, OSL uses light and TL uses heat.
Archaeomagnetism	Fired material or kilns	Records cooling after sufficient heating. Check the regional reference curve and later movement.

Relative versus numerical concerns age expression. Direct versus indirect concerns the relationship between a sample and the target event. Growth, radioactive decay, stored signals and magnetism describe different clocks. These perspectives can apply to the same measurement.

Environmental archives

Tree rings, lake varves and ice layers form through different processes. A varve is a set of sediment layers representing one year. Visible bands require checks for annual formation, continuity and dating. Ring width can also respond to age, competition or insect damage as well as climate.

Scientific dating methods: Historic England

In-class exercise: begin with A

Samples A-D are fictional exercise settings, separate from the real specimens photographed in the slides. You may refer to the method table.

A: a charred seed from a hearth

The target is activity involving the fire. Work through the following example with the instructor.

Question	Worked answer
What do we measure?	Radiocarbon in the seed.
Which event does it date?	An age related to when the plant incorporated carbon.
What else must we check?	Does the seed belong to activity at this hearth? Check contamination or mixing and calibrate the result.

B: historic timber with outer wood removed

You want to date construction or repair. What could you measure, which event would it date, and what information is missing?

C: fired material in an apparently undisturbed kiln

You want to date the last firing. Which record could you use, and what information would you need for comparison?

D: wind-blown sand buried at a site

You want to date deposition. What would start the clock, and what conditions would you check?

Checking the exercise and reading ages

Example explanations for B-D

Explain the dated event and further checks as well as naming a method. Suitable methods depend on sample conditions.

Sample	Candidate method	Meaning and checks
B: timber	Dendrochronology	Dates surviving rings. Missing outer wood prevents an immediate felling or construction date. Check for reuse.
C: fired material	Archaeomagnetism	Records final cooling after sufficient heating. Check location, the regional reference and movement after firing.
D: sand	OSL	Assesses burial time after sufficient exposure to light. Evaluate resetting and the subsequent radiation dose.

Reference years and notation

1000 ± 30 BP is an example of a conventional radiocarbon age referenced to 1950. Subtracting it from the current year is incorrect. Subtraction from 1950 does not produce a calendar age either.

Calibrated ages may use cal BP, cal BCE or cal CE. The reference year for cal BP is also 1950. Record the calibration curve, assumptions and the way uncertainty is reported. Historical CE/BCE notation has no year zero.

A calendar-dated ring does not by itself explain the cause of an event. Establish the date before interpreting environmental change or human activity.

In-class reflection

Choose one example. Write one sentence explaining what it can date, and one explaining what the date alone cannot establish.

Radiocarbon calibration: University of Oxford

After-class review and sources

Lesson 1 videos support review and independent study after class. Viewing them before the first session is not required. Select a figure and identify its sample, region, period and dating method using the explanation from class.

From figures to original sources

1. Overview of dendrochronology: Wikipedia
2. Classifying dating methods: Wikipedia
3. Dendrochronology and archaeology: National Park Service
4. Radiocarbon samples and interpretation: University of Oxford
5. Yanase sugi chronology: Yonenobu, Hoshino and Ohyama (2021)
6. Environmental archives and paleoclimate: NOAA
7. International Tree-Ring Data Bank
8. Learning crossdating: University of Arizona

Begin with the caption and check what the figure shows. Follow a DOI or reference to the original paper. Useful search terms include dendrochronology, crossdating and radiocarbon calibration.

Questions for the next lesson

Where are rings in a wood cross-section? What does the cambium produce, and how do earlywood and latewood differ? The next session introduces wood anatomy and ring structure from the basics.

This is the English handout. Japanese materials are available separately. Image sources and licences appear in the slide links and the accompanying source document.

Lesson 1 Connecting tree rings and radiocarbon

For slide 11 and Lesson 3, slide 13. A four-minute check; no preparation is required for Lesson 1.

Two clocks, connected by known-age wood

Crossdate rings to establish their calendar years. Measure their radiocarbon content. Many pairs of known calendar ages and measured radiocarbon ages provide evidence for a calibration curve. Tree rings are a major source of calibration data, but do not supply the entire curve.

For an unknown sample, use the relationship in the other direction: compare the radiocarbon measurement and its uncertainty with an appropriate curve to obtain calendar-age ranges and probabilities. Atmospheric radiocarbon has varied, so subtracting a radiocarbon age from 1950 does not by itself give a calendar year.

Question	Evidence needed
Calendar age of a ring	Crossdating against a securely dated reference
Calendar age of an unknown sample	Radiocarbon measurement, uncertainty and suitable curve
Date of an archaeological activity	Relationship between growth, use, deposition and the activity

Why measure radiocarbon in wood whose calendar age is already known? Explain the separate roles of crossdating and radiocarbon measurement.

Check

The independently known calendar age makes the ring a calibration reference. Irregularities in the curve can produce several possible calendar-age ranges for one measurement.

Historic England: radiocarbon calibration